

1. Administrative & Study Identification

Full Study Title: An Investigational Immuno-Therapy Study to Determine the Safety and Effectiveness of Nivolumab and Daratumumab in Patients With Multiple Myeloma **Official Title:** Multiple Phase 1/2 Cohorts of Nivolumab Monotherapy or Nivolumab Combination Regimens Across Relapsed/Refractory Hematologic Malignancies **Short Title/Acronym:** CheckMate 039 / CA209-039 **Protocol Number:** CA209-039 **NCT Number:** NCT01592370 **Posting ID:** 805099539458728727 **Sponsor/Company Name:** Bristol Myers Squibb **Trial Status:** COMPLETED **Investigational Product:** Nivolumab and Daratumumab (with or without Pomalidomide and Dexamethasone) **Preferred UMLS Name:** Antineoplastic Agents, Immunological **Phase of Development:** PHASE1, PHASE2 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the safety, tolerability, and side effects of the combination treatment of nivolumab and daratumumab in participants with relapsed/refractory multiple myeloma. **Secondary Objectives:** To evaluate Best Overall Response, Best Overall Response in the Multiple Myeloma Group, and Duration of Response.

2.2 Background & Rationale Multiple myeloma (MM) is largely incurable despite available therapies, and most patients eventually develop relapsed/refractory (RR) disease, representing a significant unmet medical need. Nivolumab blocks the PD-1 pathway to augment antitumor immunity, while Daratumumab targets CD38+ myeloma cells. Combining these immunotherapies with distinct mechanisms of action may overcome drug resistance, improve depth of response, and extend survival in heavily pretreated patients across relapsed/refractory hematologic malignancies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (e.g., Daratumumab monotherapy arm in specific cohorts) **Blinding:** Open-label **Randomization:** Randomized (Parallel group assignment) **Trial Status:** COMPLETED

3.2 Planned Enrollment & Duration Targeted Enrollment: 320 participants **Number of Sites:** Multicenter, International

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria** 1. Age $\geq$ 18 years with an ECOG performance status of 0 or 1. 2. Have received at least 3 prior lines of therapy, including a proteasome inhibitor (PI) and an immunomodulatory agent (IMiD), OR have disease that is double refractory to a PI and IMiD. 3. More than 12 weeks post-transplant of your own blood forming stem cells (autologous transplant). 4. Have detectable disease measured by a specific protein in your blood and/or urine. 5. Must consent to a bone marrow aspirate or biopsy.
- 4.2 Exclusion Criteria** 1. Solitary bone or extramedullary plasmacytoma as the only evidence of plasma cell dyscrasia, or monoclonal gammopathy of undetermined significance (MGUS), smoldering multiple myeloma (SMM), primary amyloidosis, Waldenstrom's macroglobulinemia, POEMS syndrome, or active plasma cell leukemia. 2. Prior therapy with anti-PD-1, anti-PD-L1, anti-PD-L2, anti-CD137, anti CTLA 4, or anti-CD38 antibody, or prior allogeneic stem cell transplantation. 3. Seropositive for human immunodeficiency virus (HIV), Hepatitis B surface antigen, or Hepatitis C antibody positive (except if HCV-RNA negative), or history of active chronic hepatitis B or C. 4. History of central nervous system involvement or symptoms suggestive of central nervous system involvement by multiple myeloma.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** Nivolumab paired with Daratumumab (with Daratumumab pre/post-infusion standard dosing depending on patient age/weight, e.g., 16-20 mg). Cohort A also received Pomalidomide and Dexamethasone. **Route of Administration:** Intravenous (IV) for Nivolumab and Daratumumab; Oral (PO) for Pomalidomide/Dexamethasone. **Duration of Treatment:** Until disease progression or unacceptable toxicity.
- 5.2 Concomitant & Prohibited Therapy Permitted:** Standard premedications to mitigate infusion-related reactions associated with monoclonal antibodies. **Prohibited:** Concurrent anti-cancer therapy, other investigational agents, or live virus vaccines during the study period.
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening
2	0 to 14	Baseline

Day 0 | Randomization, First Dose, Safety Baseline | | **Interim** |
Every 2-4 Weeks | Safety Labs, Efficacy Assessments (Blood/Urine
Proteins), Infusions | | **End of Study** | At Progression | Final
Vital Signs, Tumor Assessment, Exit Interview, Survival Follow-up
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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Number of Participants That Experienced Drug Related Grade 3-4 AEs:** Number and percent of participants that experienced drug related Grade 3-4 AEs occurring up to 100 days after the last dose of study drug. 2. **Number of Participants That Experienced Drug Related Grade 3-4 SAEs:** Number and percent of participants that experienced drug related Grade 3-4 SAEs occurring up to 100 days after the last dose of study drug. 3. **Number of Participants With Clinical Laboratory Abnormalities by Worst Toxicity Grade - Liver:** Number and percent of participants that experienced drug related Grade 3-4 AEs occurring up to 100 days after the last dose of study drug.

7.2 Secondary Endpoints 1. Best Overall Response. 2. Best Overall Response - Multiple Myeloma Group. 3. Duration of Response.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs/SAEs at each clinical visit and infusion appointment, focusing on immune-related adverse events (irAEs) and infusion reactions up to 100 days after the last dose. **Serious Adverse Events (SAEs):** Required reporting to the sponsor and regulatory authorities typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Periodic review of unblinded safety and efficacy data by an independent data monitoring committee.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for baseline characteristics and efficacy outcomes; Safety Set for all participants receiving ≥ 1 dose of the investigational product. **Sample Size Justification:** Designed to evaluate the tolerability and safety profile (Phase 1) and provide preliminary power to detect meaningful efficacy signals (Phase 2) across the 320 participants. **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier estimates) with right-censoring for patients who have not progressed or died by the data cutoff date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring visits to ensure protocol compliance, safety tracking, and proper investigational product handling. **Audit Trail:** Robust electronic case report form (eCRF) locking procedures, query resolutions, and source data verification.

11. Study Identifiers & Governance

Primary ID: CA209-039 **Registry Number:** NCT01592370 **Posting ID:** 805099539458728727 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** CheckMate 039 **Official Regulatory Title:** Multiple Phase 1/2 Cohorts of Nivolumab Monotherapy or Nivolumab Combination Regimens Across Relapsed/Refractory Hematologic Malignancies **Trial Status:** COMPLETED

12. Clinical Framework (Disease Areas)

Conditions / Disease Areas: Non-Hodgkin's Lymphoma, Hodgkin Lymphoma (Hodgkin Disease), Multiple Myeloma **Primary Condition Targeted in Expansion:** Relapsed or Refractory Multiple Myeloma (RRMM) **Diagnosis Threshold:** Detectable disease measured by specific protein in blood/urine + Bone marrow aspirate/biopsy **Medical Methodology:** Dual Immuno-Therapy (PD-1 Blockade + CD38 targeting)

13. Study Procedures & Methodology

Management Pathway: Standard oncology treatment pathway for relapsed/refractory hematologic malignancies **Education Protocol (HCP):** Site initiation visits and immuno-oncology toxicity management training **Education Protocol (Patient):** Informed consent process, reporting of immune-related side effects **Quality Audit Mechanism:** Source data verification, central laboratory calibration, and protocol adherence tracking

14. Observation & Follow-Up Schedule

Total Duration: Treatment until disease progression; Long-term follow-up for overall survival **Onsite Visit Cadence:** Every treatment cycle (typically 2-4 weeks) **Remote Monitoring Frequency:** As indicated for safety follow-up and survival tracking **Checklist Review Interval:** Disease evaluation (blood/urine markers) performed at regular intervals per protocol

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					